

Expression profile of DOT1 genes

How are DOT1 genes expressed ?

```
library(tidyverse)
library(readxl)
```

Extract DOT1 genes expression and specificity

```
Ec_DOT1.1 = "Ec-03_002870"
Ec_DOT1.2 = c("Ec-25_002640", "Ec-25_002980", "Ec-25_003090")
Ec_DOT1.3 = c("Ec-06_007110", "Ec-08_001840", "Ec-12_004580", "Ec-24_003490")
Ec_DOT1.4 = "Ec-10_005980"
Ec_DOT1.5 = "Ec-16_000810"
# save DOT1 as dataframe
Ec_DOT1_label <- tibble::tibble(
  GeneID = c(
    Ec_DOT1.1,
    Ec_DOT1.2,
    Ec_DOT1.3,
    Ec_DOT1.4,
    Ec_DOT1.5
  ),
  Labels = c(
    "DOT1.1",
    rep("DOT1.2", length(Ec_DOT1.2)),
    rep("DOT1.3", length(Ec_DOT1.3)),
    "DOT1.4",
    "DOT1.5"
  )
)
```

from the original mapping of the hourglass paper

```
selectGenes <- function(counts, min.count=2){
  keep <-
    counts[rowMeans(counts)>min.count, ]
  return(keep)
}

sample_info_ec <-
  readr::read_tsv("../Fucus_transcriptome/metadata_final.tsv") |>
  dplyr::filter(Species == "Ec") |>
  dplyr::filter(
    SeqProt == "Total" | Stage %in% c("meiospore", "mitospore") |>
    dplyr::filter(Tissue %in% c("All", "rhizoid")) |>
    tidyr::unite("stage_tissue", Stage:Tissue, remove = FALSE) |>
    dplyr::mutate(
      stage_tissue = stringr::str_remove(stage_tissue, '_All')
```

```

) |>
dplyr::arrange(
  base::match(
    Stage,
    c("meiospore", "earlyGA", "immGA", "matGA", "oldGA", "gamete", "earlyPSP", "immGA")
  ) |>
dplyr::filter(!SampleName %in% c("F_oldGA_R_1", "F_oldGA_R_2"))

sample_info_ec$Stage <-
  factor(
    sample_info_ec$Stage,
    levels = c("meiospore", "earlyGA", "immGA", "matGA", "oldGA", "gamete", "earlyPSP", "immGA")
  )

Ec_files <-
  stringr::str_c(
    "../Fucus_transcriptome/data/Ec_salmon2/",
    sample_info_ec$LibName,
    "/quant.sf" ) |>
  purrr::set_names(sample_info_ec$SampleName)

sample_info_ec.processed <-
  sample_info_ec |>
  dplyr::filter(Sex %in% c("F", "M")) |>
  tidyr::unite("sampleID", c(Sex, Stage), remove = FALSE) |>
  dplyr::select(SampleName, sampleID)

Ec_tx2gene <-
  readr::read_tsv("../Fucus_transcriptome/data/Ec_salmon2/salmon_tx2gene.tsv",
    col_names = c("txID", "GeneID")) |>
  dplyr::mutate(GeneID = stringr::str_remove(GeneID, "\\.[0-9]+$")) |>
  dplyr::select(txID, GeneID)

# use abundance
Ec_txi <-
  tximport::tximport(
    Ec_files,
    type = "salmon",
    tx2gene = Ec_tx2gene,
    countsFromAbundance = "lengthScaledTPM")

Ec_TPM_nofilt <-
  Ec_txi$counts |>
  as.data.frame() |>
  selectGenes(min.count = 0) |>
  tibble::rownames_to_column("GeneID")

# save for now
Ec_TPM_nofilt |>
  readr::write_csv("data/hourglass_data/Ec_abundance_nofilt.csv")

Ec_TPM_nofilt <-
  readr::read_csv("data/hourglass_data/Ec_abundance_nofilt.csv")

```

```

Ec_TPM_nofilt.long <-
  Ec_TPM_nofilt |>
  dplyr::left_join(Ec_DOT1_label, by = "GeneID") |>
  tidyr::pivot_longer(
    cols = -c(GeneID, Labels),
    names_to = "sample",
    values_to = "TPM"
  ) |>
  dplyr::inner_join(sample_info_ec.processed, by = c("sample" = "SampleName")) |>
  dplyr::summarise(
    TPM = mean(TPM),
    .by = c(GeneID, Labels, sampleID)
  )

Ec_TPM_nofilt.DOT1 <-
  Ec_TPM_nofilt.long |>
  dplyr::filter(GeneID %in% c(
    Ec_DOT1.1,
    Ec_DOT1.2,
    Ec_DOT1.3,
    Ec_DOT1.4,
    Ec_DOT1.5
  ))

# mean TPM across all rows
Ec_TPM_nofilt.meanTPM.DOT1 <-
  Ec_TPM_nofilt.DOT1 |>
  dplyr::summarise(meanTPM = mean(TPM), .by = c(GeneID, Labels))

# radial plot
Ec_TPM_nofilt.DOT1 |>
  # seperate by F and M
  tidyr::separate(
    sampleID,
    sep = "_",
    into = c("sex", "stage")
  ) |>
  dplyr::mutate(stage = factor(
    stage,
    levels = c(
      "meiospore",
      "immGA",
      "matGA",
      "oldGA",
      "gamete",
      "earlyPSP",
      "immPSP",
      "matPSP",
      "mitospore"
    )
  )) |>
  dplyr::mutate(GeneID = factor(GeneID, levels = Ec_DOT1_label$GeneID)) |>
  ggplot2::ggplot(aes(x = stage, y = log2(TPM+1), group = interaction(GeneID, sex), color = GeneID)) +
  ggplot2::geom_line() +

```

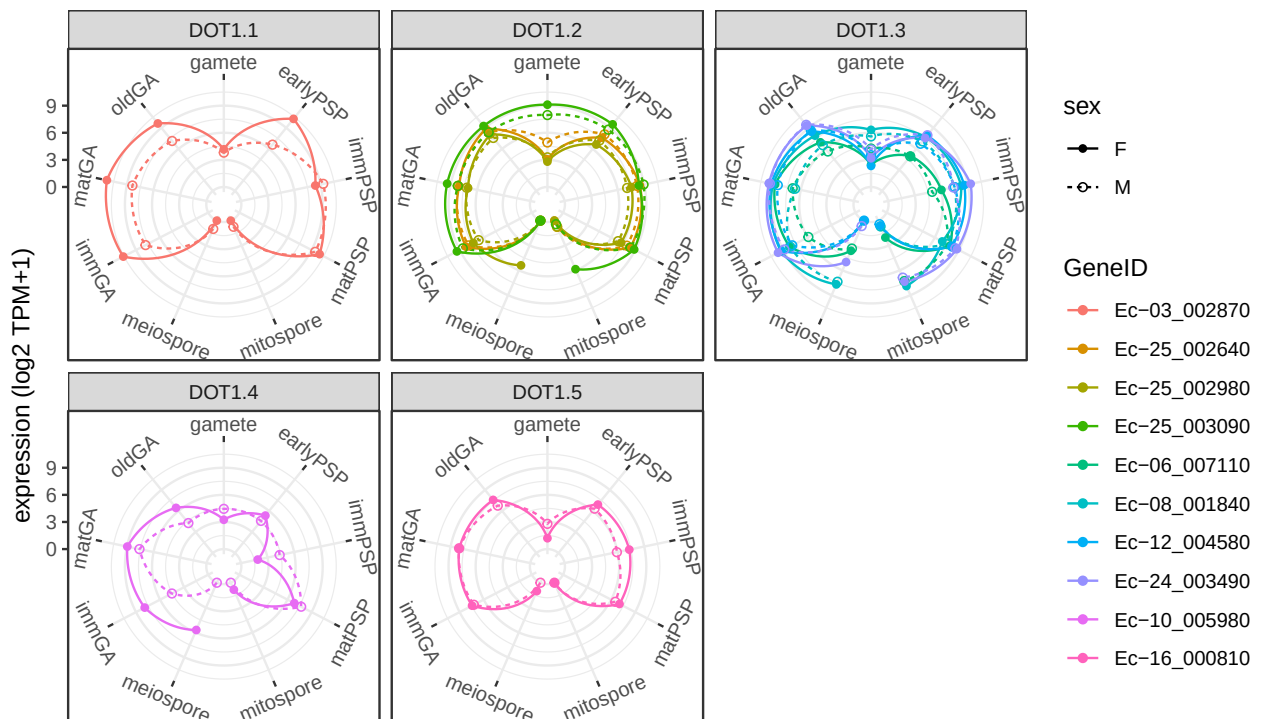
```

ggplot2::geom_point(aes(shape = sex)) +
ggplot2::facet_wrap(~Labels, nrow = 2) +
ggplot2::coord_radial(clip = "off",
                      start = -pi,
                      inner.radius = 0.1,
                      rotate.angle = TRUE) +

ggplot2::labs(
  x = "",
  y = "expression (log2 TPM+1)",
  title = "DOT1 genes expression in Ectocarpus life cycle"
) +
ggplot2::theme(
  axis.text.x = element_text(size = 7),
  # axis.text.x = element_blank(),
  legend.title = element_blank(),
  legend.position = "bottom"
  # plot.margin = margin(40, 40, 40, 40)
) +
ggplot2::guides(
  theta = guide_axis_theta(angle = 0),
  r = guide_axis(angle = 0)
) +
ggplot2::scale_shape_manual(values = c(16, 1)) +
ggplot2::theme_bw()

```

DOT1 genes expression in Ectocarpus life cycle



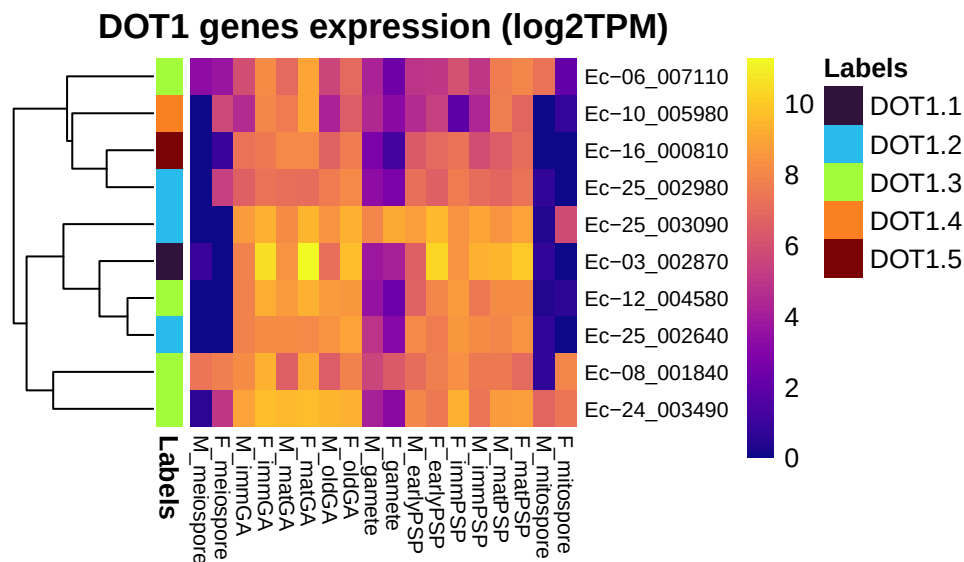
```
# ggplot2::ggsave("figure/Ec_DOT1_expression_radial_nofilt.pdf", width = 8, height = 5)
```

As heatmap

```

library(pheatmap)
Ec_TPM_nofilt.DOT1.matrix <-
  Ec_TPM_nofilt.DOT1 |>
  tidyr::pivot_wider(
    names_from = sampleID,
    values_from = TPM
  ) |>
  dplyr::select(-Labels) |>
  tibble::column_to_rownames("GeneID") |>
  as.matrix()
pheatmap::pheatmap(
  log2(Ec_TPM_nofilt.DOT1.matrix+1),
  # colour by DOT1 type
  annotation_row = Ec_DOT1_label |> tibble::column_to_rownames("GeneID"),
  clustering_method = "complete",
  cluster_cols = FALSE,
  annotation_colors = list(
    Labels = viridis::turbo(n = 5) |>
      setNames(nm = unique(Ec_DOT1_label$Labels))
  ),
  fontsize_row = 8,
  fontsize_col = 8,
  main = "DOT1 genes expression (log2TPM)",
  border_color = NA,
  color = viridis::viridis(100, option = "plasma")
) -> heatmap_fig

```



```

# save heatmap_fig
# ggplot2::ggsave("figure/Ec_DOT1_expression_heatmap_nofilt.pdf", plot = heatmap_fig, width = 5, height

```

As heatmap only for separate sex

```

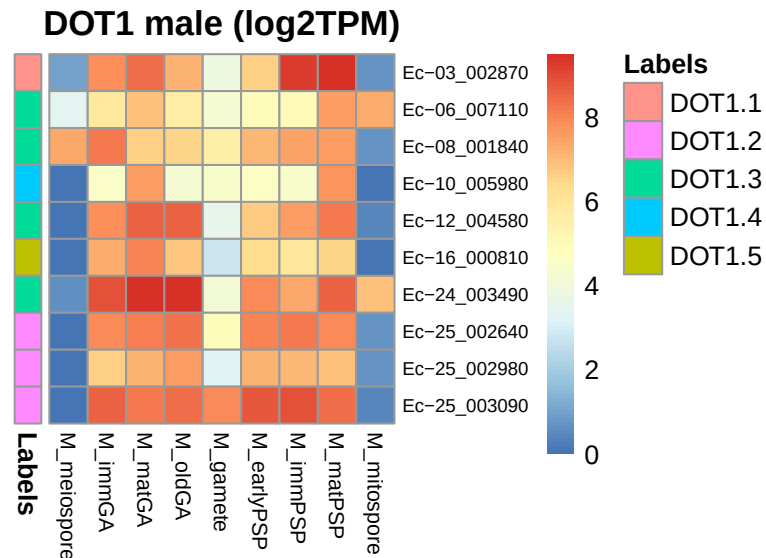
library(pheatmap)
Ec_TPM_nofilt.DOT1.matrix.M <-
  Ec_TPM_nofilt.DOT1 |>
  dplyr::filter(stringr::str_detect(sampleID, "^M_")) |>

```

```

tidyr::pivot_wider(
  names_from = sampleID,
  values_from = TPM
) |>
dplyr::select(-Labels) |>
tibble::column_to_rownames("GeneID") |>
as.matrix()
pheatmap::pheatmap(
  log2(Ec_TPM_nofilt.DOT1.matrix.M+1),
  # colour by DOT1 type
  annotation_row = Ec_DOT1_label |> tibble::column_to_rownames("GeneID"),
  clustering_method = "complete",
  cluster_cols = FALSE,
  cluster_rows = FALSE,
  fontsize_row = 7,
  fontsize_col = 8,
  main = "DOT1 male (log2TPM)"
)

```



```

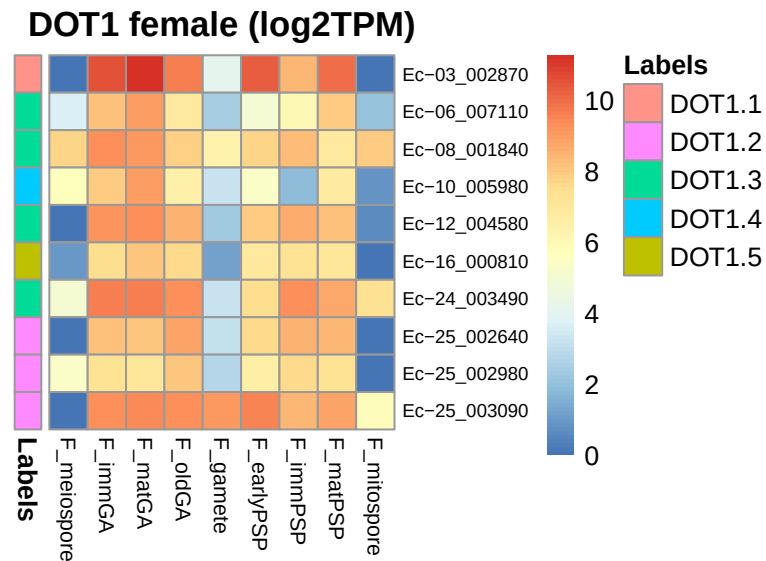
Ec_TPM_nofilt.DOT1.matrix.F <-
  Ec_TPM_nofilt.DOT1 |>
  dplyr::filter(stringr::str_detect(sampleID, "^F_")) |>
  tidyr::pivot_wider(
    names_from = sampleID,
    values_from = TPM
  ) |>
  dplyr::select(-Labels) |>
  tibble::column_to_rownames("GeneID") |>
  as.matrix()
pheatmap::pheatmap(
  log2(Ec_TPM_nofilt.DOT1.matrix.F+1),
  # colour by DOT1 type
  annotation_row = Ec_DOT1_label |> tibble::column_to_rownames("GeneID"),
  clustering_method = "complete",
  cluster_cols = FALSE,
  cluster_rows = FALSE,

```

```

    fontsize_row = 7,
    fontsize_col = 8,
    main = "DOT1 female (log2TPM)"
  )

```



As line plot

```

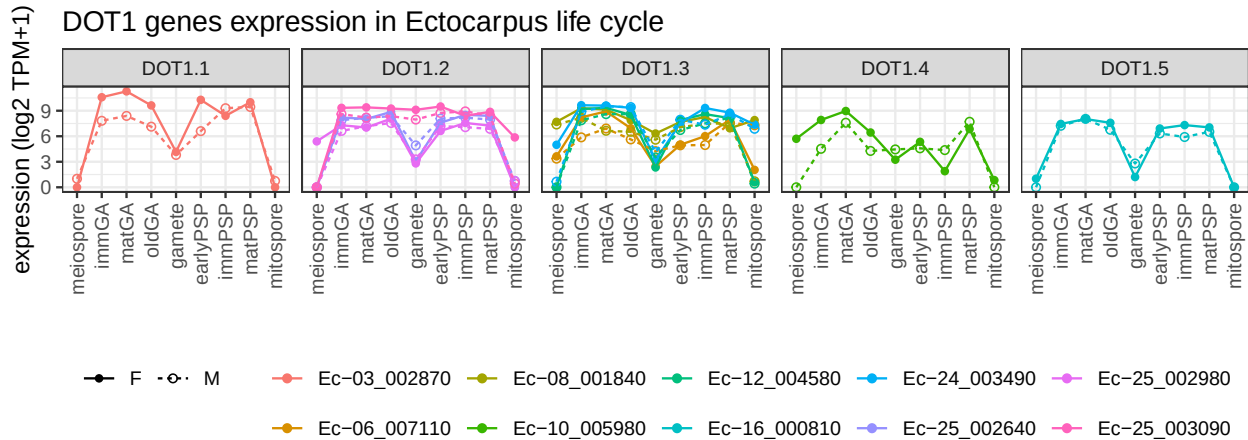
# line plot
Ec_TPM_nofilt.DOT1 |>
  # seperate by F and M
  tidyr::separate(
    sampleID,
    sep = "_",
    into = c("sex", "stage")
  ) |>
  dplyr::mutate(stage = factor(
    stage,
    levels = c(
      "meiospore",
      "immGA",
      "matGA",
      "oldGA",
      "gamete",
      "earlyPSP",
      "immPSP",
      "matPSP",
      "mitospore"
    )
  )) |>
  ggplot2::ggplot(aes(x = stage, y = log2(TPM+1), group = interaction(GeneID, sex), color = GeneID)) +
  ggplot2::geom_point(aes(shape = sex)) +
  ggplot2::geom_line() +
  ggplot2::facet_grid(~Labels) +
  ggplot2::theme_bw() +
  ggplot2::labs(
    x = "",

```

```

y = "expression (log2 TPM+1)",
title = "DOT1 genes expression in Ectocarpus life cycle"
) +
ggplot2::scale_shape_manual(values = c(16, 1)) +
ggplot2::theme(
  legend.title = element_blank(),
  legend.position = "bottom",
  # 90 degrees x axis text
  axis.text.x = element_text(angle = 90, hjust = 1, vjust = 0.5)
)

```



Compute tau with the nofilter data

```

# check Kryuchkova-Mostacci & Robinson-Rechavi for more functions to add.
specificity <- function(abundance, measure = "tau"){
  abundance_longer <-
    abundance |>
    duckplyr::as_duckdb_tibble() |>
    dplyr::select(1, where(is.numeric)) |>
    tidyr::pivot_longer(!1, names_to = "sample", values_to = "TPM")
  if(measure == "gini"){
    breadth_out <- abundance_longer |>
    dplyr::summarise(
      gini = DescTools::Gini(TPM),
      .by = GeneID
    )
  } else if(measure == "tau"){
    n_col <- ncol(abundance) - 1
    breadth_out <- abundance_longer |>
    dplyr::summarise(
      tau = sum(1 - (TPM/max(TPM)))/(n_col-1),
      .by = GeneID
    )
  } else if(measure == "H"){
    breadth_out <- abundance_longer |>
    dplyr::summarise(
      H = DescTools::Entropy(TPM/sum(TPM)),
      .by = GeneID
    )
  } else if(measure == "rH"){

```



```

# KL divergence from uniform distribution
n_col <- ncol(abundance) - 1
x_1 <-
breadth_out <- abundance_longer |>
  dplyr::mutate(
    # rH = entropy::KL.empirical(y1 = TPM/sum(TPM), y2 = 1/n_col),
    y1 = TPM/sum(TPM),
    y2 = 1/n_col,
    .by = GeneID
  ) |>
  dplyr::summarise(
    rH = entropy::KL.empirical(y1, y2),
    .by = GeneID
  )
} else if(measure == "mean"){
  # not really a specificity but useful nonetheless
  breadth_out <- abundance_longer |>
    dplyr::summarise(
      meanTPM = mean(TPM),
      .by = GeneID
    )
} else {
  stop("Specify measure: \"gini\", \"tau\", \"H\", \"rH\", \"mean\"")
}
return(breadth_out)
}

Ec_TPM_nofilt.wide <-
  Ec_TPM_nofilt.long |>
  tidyr::pivot_wider(
    names_from = sampleID,
    values_from = TPM
  ) |>
  dplyr::select(-Labels)
Ec_TPM_nofilt.wide.log2 <-
  Ec_TPM_nofilt.wide |>
  dplyr::mutate(across(-GeneID, ~ log2(.x + 1)))
Ec_tau_nofilt <- specificity(Ec_TPM_nofilt.wide.log2, measure = "tau") |>
  tidyr::drop_na()
Ec_DOT1_tau_nofilt <-
  Ec_tau_nofilt |>
  dplyr::filter(stringr::str_detect(GeneID, "^Ec-(03_002970|25_002640|25_002980|25_003090|06_0071|06_0072|06_0073|06_0074|06_0075|06_0076|06_0077|06_0078|06_0079|06_0080|06_0081|06_0082|06_0083|06_0084|06_0085|06_0086|06_0087|06_0088|06_0089|06_0090|06_0091|06_0092|06_0093|06_0094|06_0095|06_0096|06_0097|06_0098|06_0099|06_0100|06_0101|06_0102|06_0103|06_0104|06_0105|06_0106|06_0107|06_0108|06_0109|06_0110|06_0111|06_0112|06_0113|06_0114|06_0115|06_0116|06_0117|06_0118|06_0119|06_0120|06_0121|06_0122|06_0123|06_0124|06_0125|06_0126|06_0127|06_0128|06_0129|06_0130|06_0131|06_0132|06_0133|06_0134|06_0135|06_0136|06_0137|06_0138|06_0139|06_0140|06_0141|06_0142|06_0143|06_0144|06_0145|06_0146|06_0147|06_0148|06_0149|06_0150|06_0151|06_0152|06_0153|06_0154|06_0155|06_0156|06_0157|06_0158|06_0159|06_0160|06_0161|06_0162|06_0163|06_0164|06_0165|06_0166|06_0167|06_0168|06_0169|06_0170|06_0171|06_0172|06_0173|06_0174|06_0175|06_0176|06_0177|06_0178|06_0179|06_0180|06_0181|06_0182|06_0183|06_0184|06_0185|06_0186|06_0187|06_0188|06_0189|06_0190|06_0191|06_0192|06_0193|06_0194|06_0195|06_0196|06_0197|06_0198|06_0199|06_0200|06_0201|06_0202|06_0203|06_0204|06_0205|06_0206|06_0207|06_0208|06_0209|06_0210|06_0211|06_0212|06_0213|06_0214|06_0215|06_0216|06_0217|06_0218|06_0219|06_0220|06_0221|06_0222|06_0223|06_0224|06_0225|06_0226|06_0227|06_0228|06_0229|06_0230|06_0231|06_0232|06_0233|06_0234|06_0235|06_0236|06_0237|06_0238|06_0239|06_0240|06_0241|06_0242|06_0243|06_0244|06_0245|06_0246|06_0247|06_0248|06_0249|06_0250|06_0251|06_0252|06_0253|06_0254|06_0255|06_0256|06_0257|06_0258|06_0259|06_0260|06_0261|06_0262|06_0263|06_0264|06_0265|06_0266|06_0267|06_0268|06_0269|06_0270|06_0271|06_0272|06_0273|06_0274|06_0275|06_0276|06_0277|06_0278|06_0279|06_0280|06_0281|06_0282|06_0283|06_0284|06_0285|06_0286|06_0287|06_0288|06_0289|06_0290|06_0291|06_0292|06_0293|06_0294|06_0295|06_0296|06_0297|06_0298|06_0299|06_0300|06_0301|06_0302|06_0303|06_0304|06_0305|06_0306|06_0307|06_0308|06_0309|06_0310|06_0311|06_0312|06_0313|06_0314|06_0315|06_0316|06_0317|06_0318|06_0319|06_0320|06_0321|06_0322|06_0323|06_0324|06_0325|06_0326|06_0327|06_0328|06_0329|06_0330|06_0331|06_0332|06_0333|06_0334|06_0335|06_0336|06_0337|06_0338|06_0339|06_0340|06_0341|06_0342|06_0343|06_0344|06_0345|06_0346|06_0347|06_0348|06_0349|06_0350|06_0351|06_0352|06_0353|06_0354|06_0355|06_0356|06_0357|06_0358|06_0359|06_0360|06_0361|06_0362|06_0363|06_0364|06_0365|06_0366|06_0367|06_0368|06_0369|06_0370|06_0371|06_0372|06_0373|06_0374|06_0375|06_0376|06_0377|06_0378|06_0379|06_0380|06_0381|06_0382|06_0383|06_0384|06_0385|06_0386|06_0387|06_0388|06_0389|06_0390|06_0391|06_0392|06_0393|06_0394|06_0395|06_0396|06_0397|06_0398|06_0399|06_0400|06_0401|06_0402|06_0403|06_0404|06_0405|06_0406|06_0407|06_0408|06_0409|06_0410|06_0411|06_0412|06_0413|06_0414|06_0415|06_0416|06_0417|06_0418|06_0419|06_0420|06_0421|06_0422|06_0423|06_0424|06_0425|06_0426|06_0427|06_0428|06_0429|06_0430|06_0431|06_0432|06_0433|06_0434|06_0435|06_0436|06_0437|06_0438|06_0439|06_0440|06_0441|06_0442|06_0443|06_0444|06_0445|06_0446|06_0447|06_0448|06_0449|06_0450|06_0451|06_0452|06_0453|06_0454|06_0455|06_0456|06_0457|06_0458|06_0459|06_0460|06_0461|06_0462|06_0463|06_0464|06_0465|06_0466|06_0467|06_0468|06_0469|06_0470|06_0471|06_0472|06_0473|06_0474|06_0475|06_0476|06_0477|06_0478|06_0479|06_0480|06_0481|06_0482|06_0483|06_0484|06_0485|06_0486|06_0487|06_0488|06_0489|06_0490|06_0491|06_0492|06_0493|06_0494|06_0495|06_0496|06_0497|06_0498|06_0499|06_0500|06_0501|06_0502|06_0503|06_0504|06_0505|06_0506|06_0507|06_0508|06_0509|06_0510|06_0511|06_0512|06_0513|06_0514|06_0515|06_0516|06_0517|06_0518|06_0519|06_0520|06_0
```



```

dplyr::left_join(Ec_DOT1_tau.nofilt) |>
# boxplot between DOT1 and rest (not caring about DOT1.[0-9])
ggplot(aes(x = ifelse(is.na(Labels), "Other genes", "DOT1 genes"), y = meanTPM)) +
geom_boxplot(outlier.shape = NA) +
# only geom jitter for labels (DOT1 genes)
geom_jitter(
  data = Ec_DOT1_mean.nofilt,
  aes(x = ifelse(is.na(Labels), "Other genes", "DOT1 genes"), y = meanTPM, color = Labels,
    width = 0.2,
    height = 0,
    size = 2,
    alpha = 0.7
) +
theme_classic() +
ggpubr::stat_compare_means(
  aes(group = ifelse(is.na(Labels), "Other genes", "DOT1 genes")),
  label = "p.format",
  label.y.npc = 0.95
) +
ggplot2::theme(
  # 45 degrees x axis text
  axis.text.x = element_text(angle = 45, hjust = 1)
) +
ggplot2::labs(
  x = "",
  y = "mean expression (log2 TPM+1)"
)

```

